- $\boxed{\text{Forza gravitazionale (massa)}}$
- Forza elettromagnetica (carica)
- Forza debole : $n \rightarrow p^+ + e^- + \nu_e$

} F. elettrodebole

- Forza forte

$$\boxed{\bar{F}(\text{parametri}) = m\bar{a}}$$

$$\Uparrow$$

$$\bar{F} = 0 \Rightarrow \bar{a} = 0 \Rightarrow \boxed{\bar{v} = \text{cost}}$$

$$\bar{F} = \text{cost} \Rightarrow \bar{a} = \text{cost}$$ ← equazione vettoriale

$$\bar{F} = \bar{F}(t) \Rightarrow \bar{a} = \bar{a}(t)$$

$$\bar{F}(t) = m\bar{a} = m\bar{a}_T + m\bar{a}_N = m\frac{dv}{dt}\bar{u}_T + m\frac{v^2}{R}\bar{u}_N =$$

$$= \bar{F}_T + \bar{F}_N$$

$$\bar{F}_T = m\frac{dv}{dt}\bar{u}_T$$ ← forza tangenziale

$$\bar{F}_N = m\frac{v^2}{R}\bar{u}_N$$ ← forza centripeta

Forza peso $\bar{P} = m\bar{a}$ $\bar{a} = \bar{g}$

$\bar{g} \downarrow$ $g = |\bar{g}| = 9.81 \ m/s^2$

$$\boxed{\bar{P} = m\bar{g}}$$

peso / massa

$P \rightarrow$ forza (N)

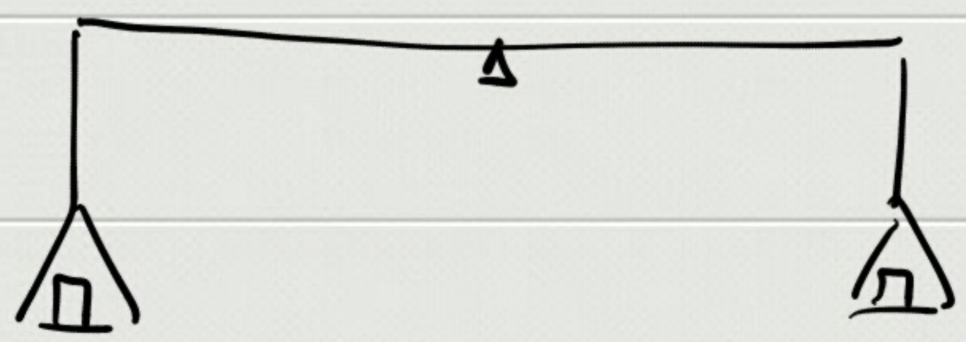
$m \rightarrow$ massa (kg)

$P \propto m$

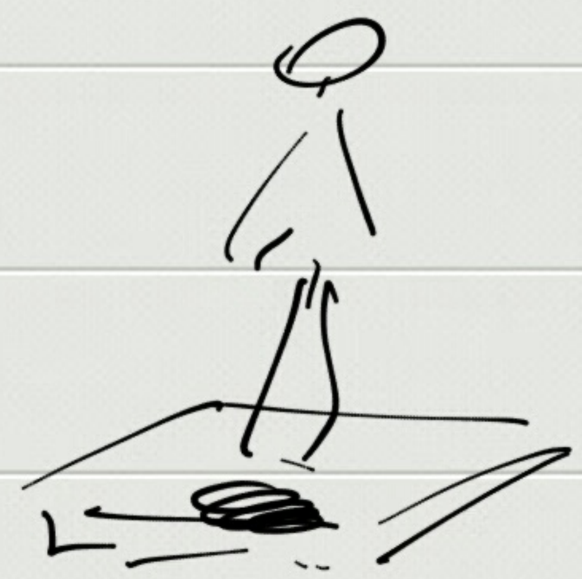
$\underline{\underline{m}} = 1 kg$ $g = 9.81 \ m/s^2$ $\Rightarrow$ $P = 9.8 \ N$

$1 \ kgf = 1 \ kgp = 9.8 \ N$

$= 1 \ kgf / 1 \ kgp$

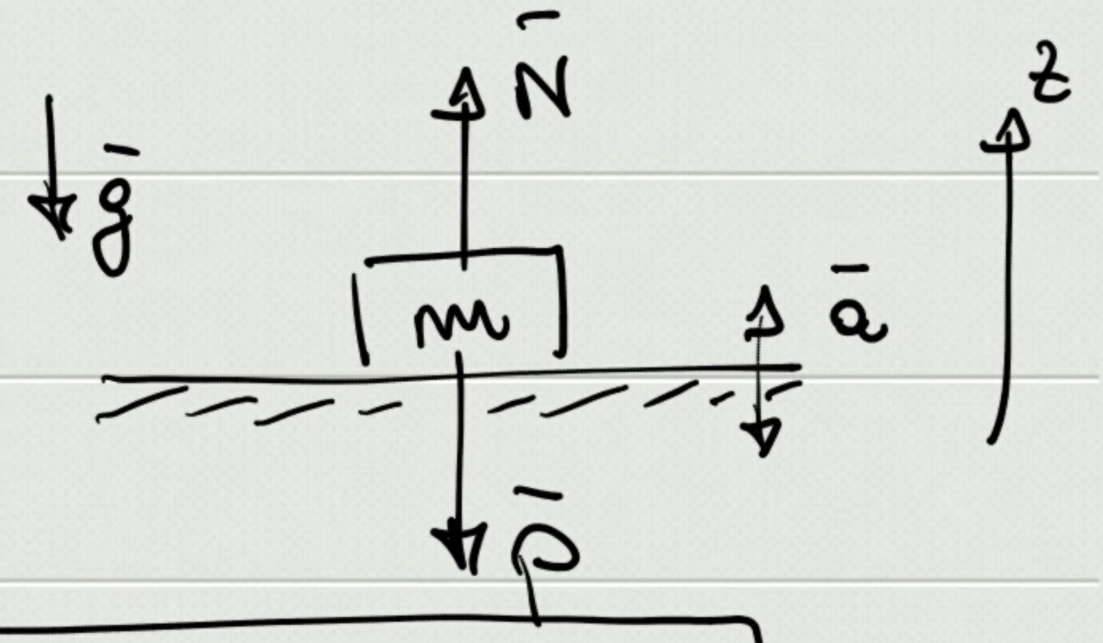


misura massa



peso (forza)

$\bar{P} = m\bar{g}$ $\qquad \bar{g} = -g\,\bar{u}_z$



$$\bar{P} + \bar{N} = m\bar{a} \Rightarrow \boxed{\bar{N} = m\bar{a} - \bar{P} = m(\bar{a} - \bar{g})}$$

- $a = 0 \Rightarrow \boxed{\bar{N}_0 = -m\bar{g}}$ $\qquad \uparrow \bar{N}_0$

- $\bar{a} = -a\,\bar{u}_z \downarrow \bar{a} \quad (a > 0) \qquad a < g$

$$\Rightarrow \bar{N}_1 = m(-a\,\bar{u}_z + g\,\bar{u}_z) = m(\overset{0}{\overset{\wedge}{g - a}})\,\bar{u}_z \qquad \uparrow \bar{N}_1$$

$$|\bar{N}_1| < |\bar{N}_0|$$

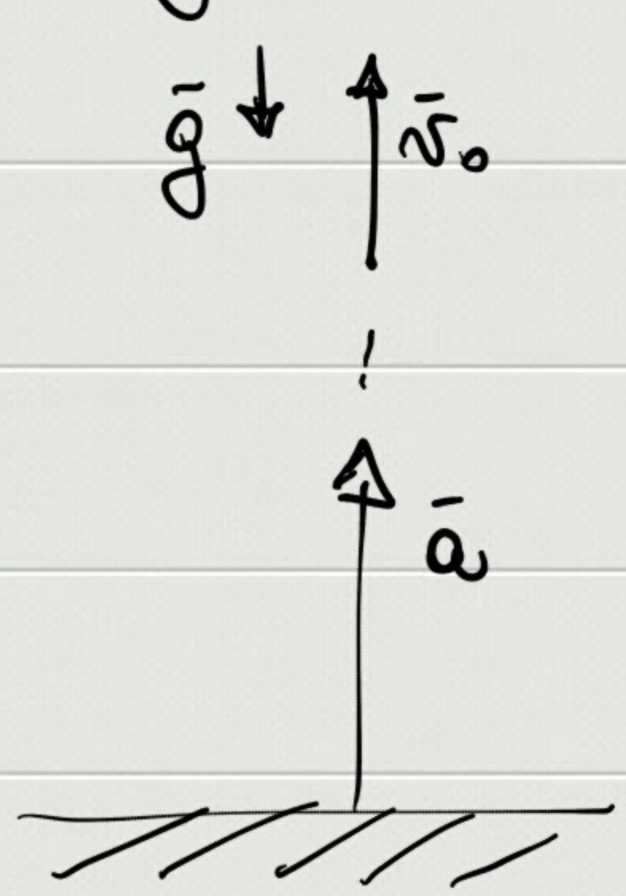
- $\bar{a} = a\,\bar{u}_z \uparrow \bar{a} \quad (a > 0)$

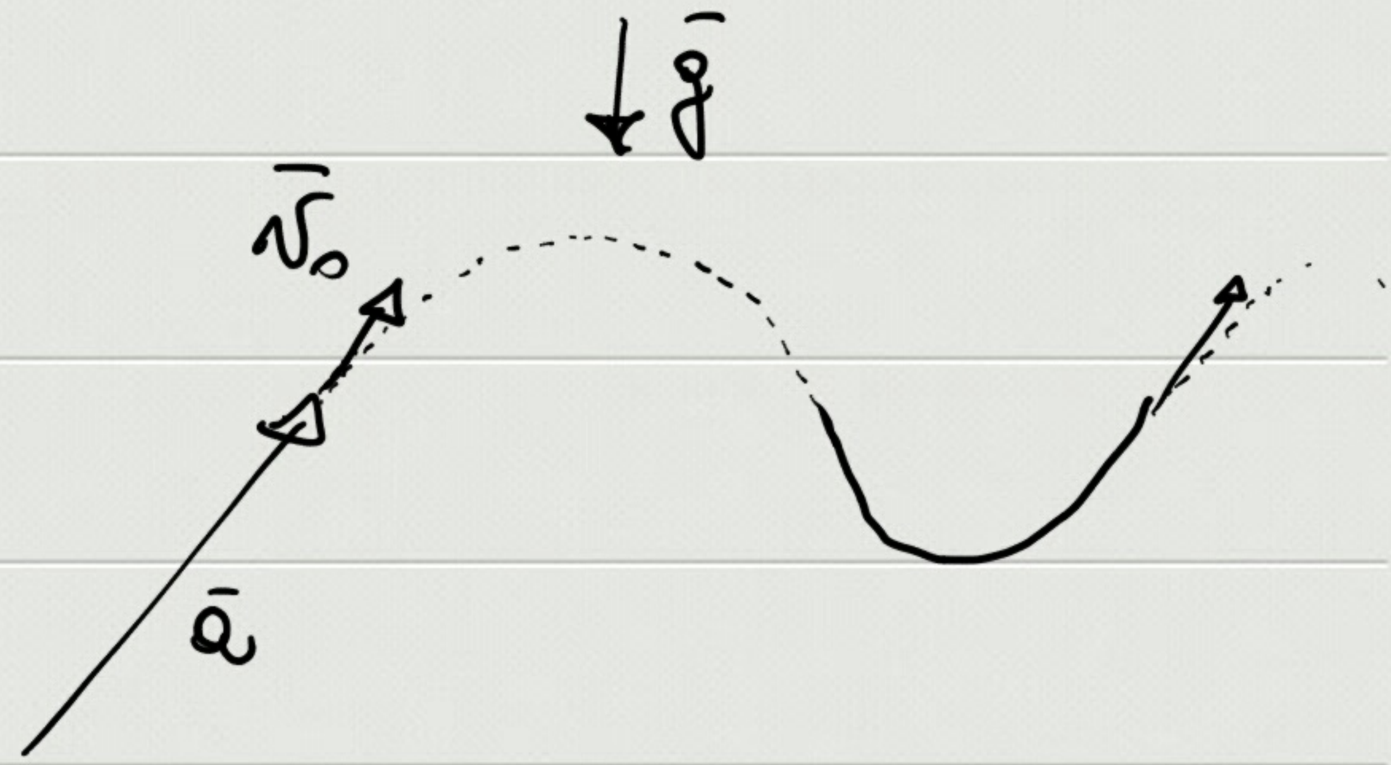
$$\Rightarrow \bar{N}_2 = m(a\,\bar{u}_z + g\,\bar{u}_z) = m(\overset{0}{\overset{\wedge}{g + a}})\,\bar{u}_z \qquad \uparrow \bar{N}_2$$

$$|\bar{N}_2| > |\bar{N}_0|$$

$N \rightarrow$ sensazione di peso!

- $\bar{a} = \bar{g} \quad \Rightarrow \quad \bar{N}_3 = 0$





- $\bar{a} = -a\,\bar{u}_z \qquad a > \bar{g}$

$\downarrow \bar{a}$

$$\bar{N}_4 = m\left(\cancel{-a\,\bar{u}_z + g\,\bar{u}_z}\right) = m\,(-a+g)\,\bar{u}_z$$

0

$\downarrow \bar{N}_4$